

EE236: Electronic Devices Lab

Lab 6: Schottky Diode IV Characterization and Transient Analysis

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1 Aim of the experiment

- To plot the forward and reverse bias I/V characteristics of 2 fabricated metal-semiconductor junction diodes (one Schottky contact and another Ohmic contact) using probe station.
- To plot the forward and reverse bias I/V characteristics of the given packaged Schottky diode
- To obtain the reverse recovery times of a regular PN junction diode and a Schottky diode and compare the two.

2 Theory

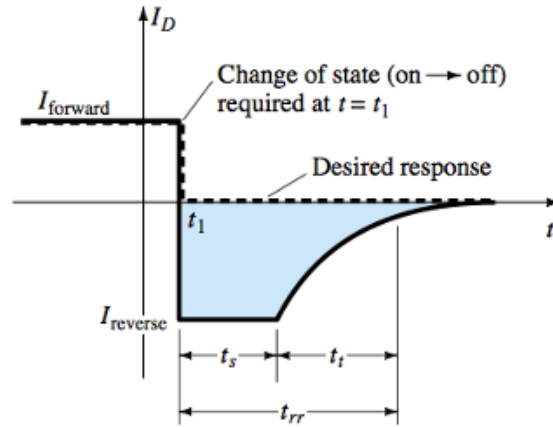
Unlike the conventional PN junction diode which is formed from a piece of P-type material and a piece of N-type material, the Schottky Diode is constructed using a metal and an N-type (or P-type) semiconductor.

Because of this, a Schottky diode has very thin depletion region. As a consequence of that, the turn on voltage of a Schottky diode is typically lower than that of a PN junction diode

When a diode is switched from forward to reverse bias, ideally it should switch instantly and block any current flow. But in reality, a reverse current

will flow for a short period of time. This is caused due to stored charge in the junction, which takes time to recombine.

Reverse recovery time of the diode is the time gap between the instant at which the reverse current starts to flow through the diode to the time instant at which it reaches to 0 (or some predefined value close to 0.)

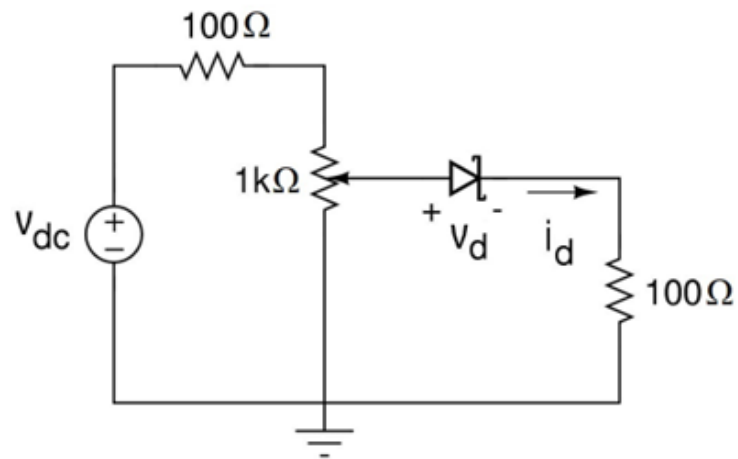


I_{rr} is the peak reverse current. Here, the RRT is measured till when the reverse current reaches 25% of I_{rr} .

3 Experiment

3.1 Part 1: I-V characterization of the fabricated diode samples

3.1.1 Circuit



3.2 S24 Sample: Readings

Voltage (V)	Current (A)
-2.0	-6 μ A
-1.8	-5 μ A
-1.6	-5 μ A
-1.4	-5 μ A
-1.2	-4 μ A
-1.0	-4 μ A
-0.8	-4 μ A
-0.6	-3 μ A
-0.4	-2 μ A
-0.2	0 μ A
0	0 mA
0.1	0.01 mA
0.2	0.01 mA
0.3	0.1 mA
0.4	0.55 mA
0.5	1.29 mA
0.6	2.1 mA
0.7	2.92 mA
0.8	3.79 mA
0.9	4.78 mA
1.0	5.74 mA
1.1	6.53 mA
1.2	7.55 mA
1.3	8.53 mA
1.4	9.44 mA
1.5	10.38 mA
1.6	11.3 mA
1.7	12.33 mA
1.8	13.3 mA

Table 1: Voltage vs Current Data with Micro and Milli Amperes

3.2.1 N15 Sample: Readings

Voltage (V)	Current (μA)
-3	-47 μA
-2.5	-41 μA
-2	-37 μA
-1.5	-28 μA
-1	-22 μA
-0.8	-20 μA
-0.6	-17 μA
-0.4	-14 μA
-0.2	-11 μA
-0.1	-10 μA
-0.05	-8 μA
-0.1	-7 μA
0	-6 μA

Voltage (V)	Current (mA)
0	-0.01 mA
0.2	0.02 mA
0.3	0.13 mA
0.4	0.24 mA
0.5	0.39 mA
0.6	0.5 mA
0.7	0.6 mA
0.8	0.71 mA
0.9	0.82 mA
1	0.91 mA
1.1	1 mA
1.2	1.1 mA
1.3	1.19 mA
1.4	1.28 mA
1.5	1.37 mA
1.6	1.46 mA
1.7	1.54 mA
1.8	1.62 mA
1.9	1.71 mA
2	1.8 mA
2.1	1.88 mA
2.2	1.95 mA
2.3	2.03 mA
2.4	2.14 mA
2.5	2.21 mA
2.6	2.3 mA
2.7	2.37 mA
2.8	2.44 mA
2.9	2.53 mA
3	2.61 mA
3.25	2.8 mA
3.5	3 mA
3.75	3.18 mA
3.94	3.34 mA

Table 2: Voltage vs Current Data

3.2.2 Results and Inferences

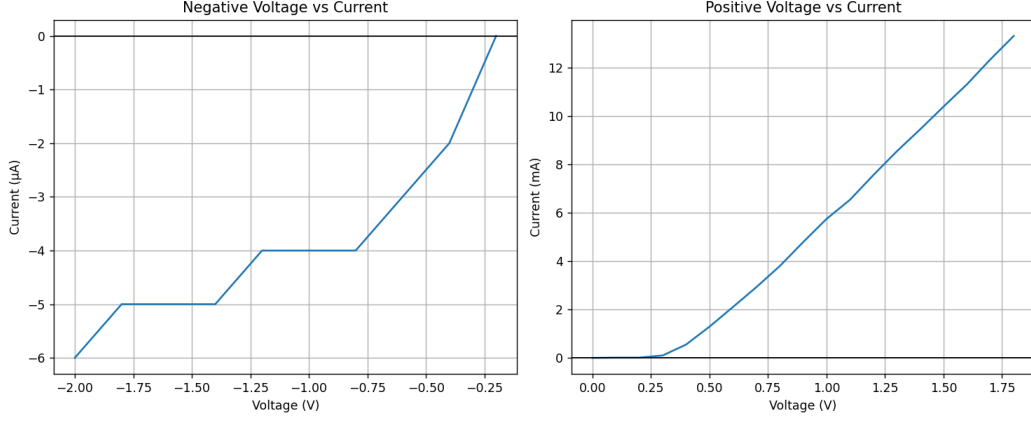


Figure 1: I-V plot for S24

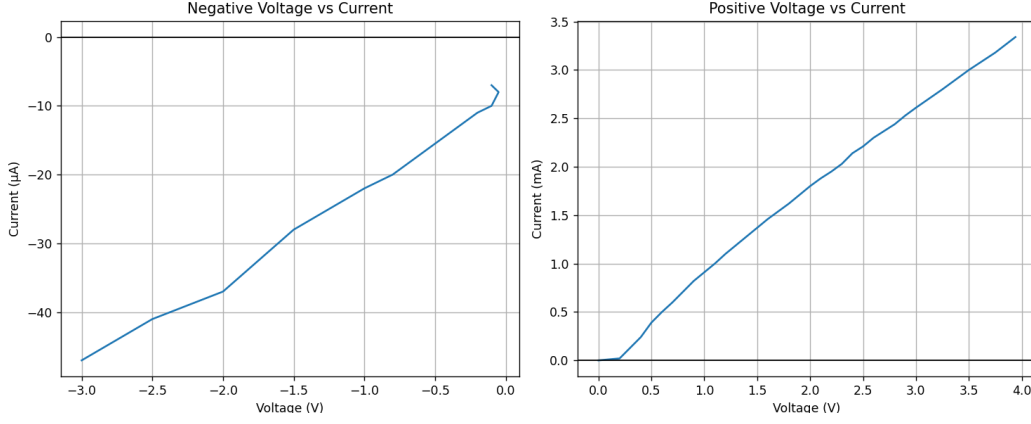


Figure 2: I-V plot for N15

As is evident from the plots, the S24 sample is Schottky and the second is N15 Ohmic contact. Ohmic contact has higher doping to give a thinner barrier, therefore it's doping is 10^{-19} cm^{-3} . For the Schottky sample, we can find the V_{bi} from the plot and the depletion width using the formula:

$$W = \left[\frac{2\epsilon_{Si}}{qN_d} (V_{bi} - V_A) \right]^{1/2} \quad (1)$$

Bias Condition	Depletion Width (W)	Built-in Voltage (V_{bi})
No Bias	762 μm	0.45 V
Forward Bias (0.5 V)	0 μm	0.45 V

Table 3: Depletion Width and Built-in Voltage under Different Bias Conditions

3.3 Part 2a: 1N5822 Schottky diode: Readings

3.3.1 Readings

Voltage (V)	Current (mA)
0.0	0.00
0.1	0.09
0.11	0.14
0.12	0.19
0.13	0.31
0.14	0.48
0.15	0.74
0.16	0.97
0.17	1.33
0.18	1.99
0.19	2.88
0.20	4.09
0.21	5.78
0.22	7.85
0.23	10.63
0.24	14.52
0.25	19.18

Table 4: Voltage vs Current for forward bias

Voltage (V)	Current (μA)
-4.0	-40
-3.5	-39
-3.0	-36
-2.5	-34
-2.0	-32
-1.5	-30
-1.0	-27
-0.5	-24
-0.2	-22
-0.1	-21
-0.04	-16
-0.01	-7
0.0	0

Table 5: Voltage vs Current for reverse bias

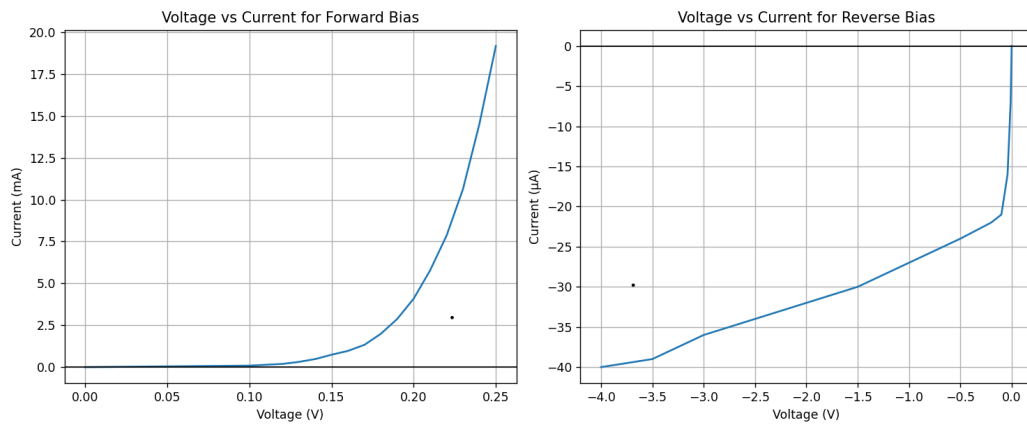
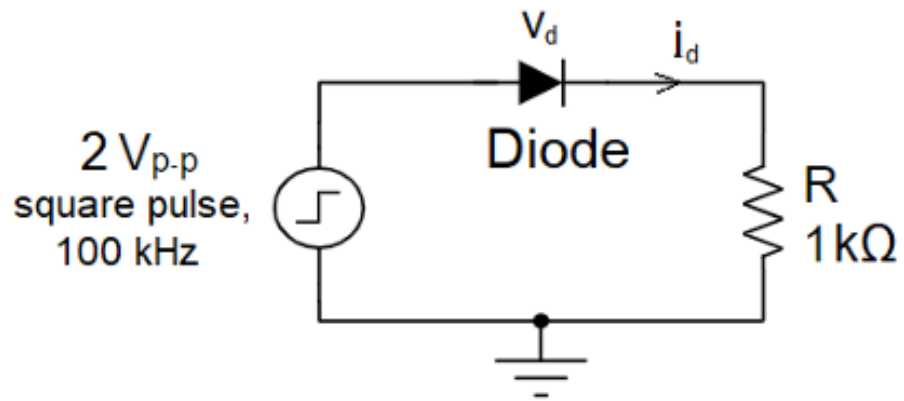


Figure 3: I-V plots

3.4 Part 2b: Measurement of RRT

3.4.1 Circuit



3.4.2 Results

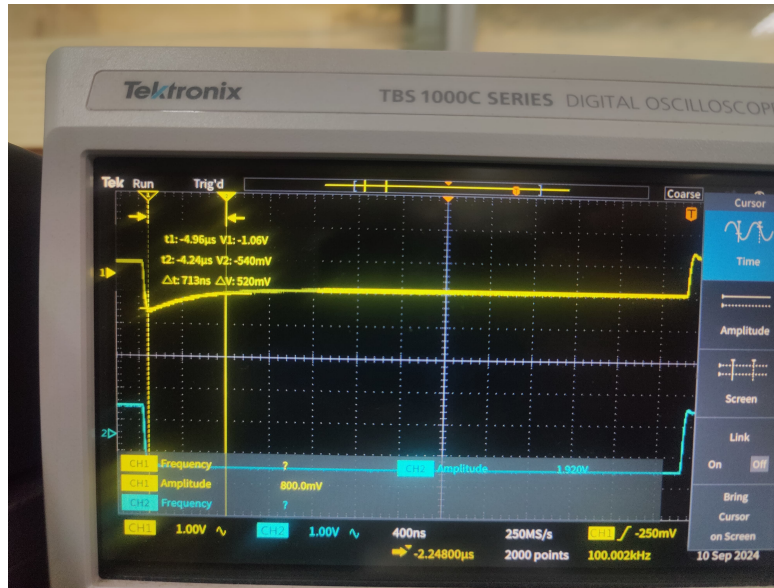


Figure 4: RRT for Schottky diode

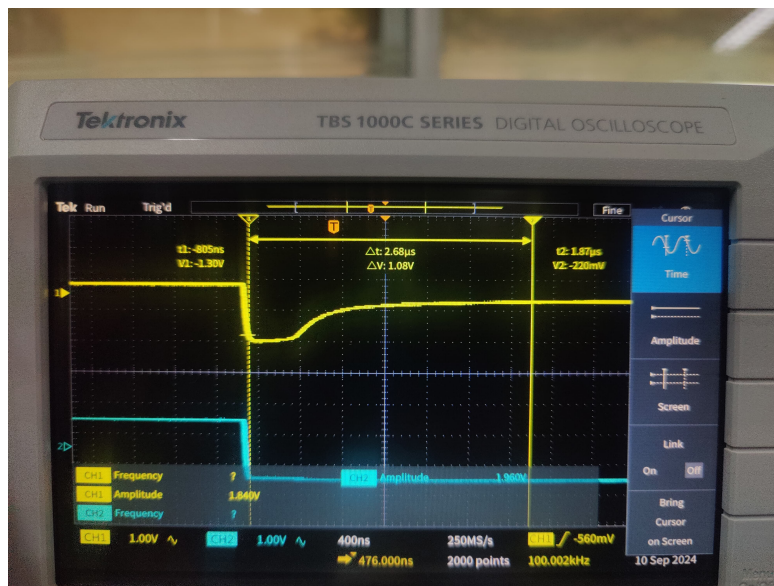


Figure 5: RRT for PN diode

Diode Type	Reverse Recovery Time
Schottky Diode	713 ns
p-n Junction Diode	2.68 μ s

Table 6: Reverse Recovery Times for Schottky and p-n Junction Diodes

The Schottky diode has a much faster reverse recovery time because it only involves majority carriers, with minimal charge storage, while the PN diode involves both majority and minority carriers, leading to slower recombination and a longer recovery time.